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Challenges in bored tunnelling beneath an earth retaining wall and canal on Singapore's Cross Island Line

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ABSTRACT: This paper presents the engineering solutions implemented during bored tunnelling works on Singapore's Cross Island Line (CRL), specifically along the alignment between Tampines North and Pasir Ris stations. The tunnel alignment encountered a challenging section where the tunnel boring machines (TBM) were required to undercross an earth retaining wall founded on bored piles, immediately followed by a stormwater canal supported by timber piles. The presence of bored piles conflicting with the tunnel alignment, combined with a narrow worksite corridor, presented complex engineering challenges. The paper details how the geology and constrained worksite space influenced the selection of foundation investigation methods and viable pile removal techniques. A comprehensive discussion of the measures implemented during pile removal and bored tunnelling is presented, along with an examination of the associated risks to existing structures. The paper also analyses the TBM performance during bored tunnelling operations at this challenging section.

1 INTRODUCTION

1.1 *Cross Island Line (CRL)*

The Cross Island Line (CRL) is Singapore's eighth Mass Rapid Transit (MRT) and a cornerstone of the nation's long-term public transport strategy. It supports the Land Transport Master Plan goal of having 80% of households within a 10-minute walk of an MRT station by the early-2030s. With its alignment exceeding 50km in length and serving 21 stations, the CRL is designed to enhance east-west connectivity and strengthen overall resilience of Singapore's MRT network.

Developed progressively across multiple construction phases, the line traverses a densely developed urban landscape, intersecting residential estates, commercial hubs and industrial zones under a wide range of geological conditions. Due to its proximity to existing people-centric infrastructure, CRL's design and construction philosophy therefore prioritises public safety, minimisation of ground improvement, and adaptability to ground variability. These guiding principles ensure that construction works can proceed safely within the tight spatial constraints of Singapore's urban setting while maintaining uninterrupted daily operations at the surface.

1.2 *Contract CR109 – Tampines north station and bored tunnels*

This paper focuses on engineering challenges at Contract CR109 of the CRL, for the design and construction of Tampines North Station and twin bored tunnels located at eastern Singapore. The overall project scope is illustrated in Figure 1.

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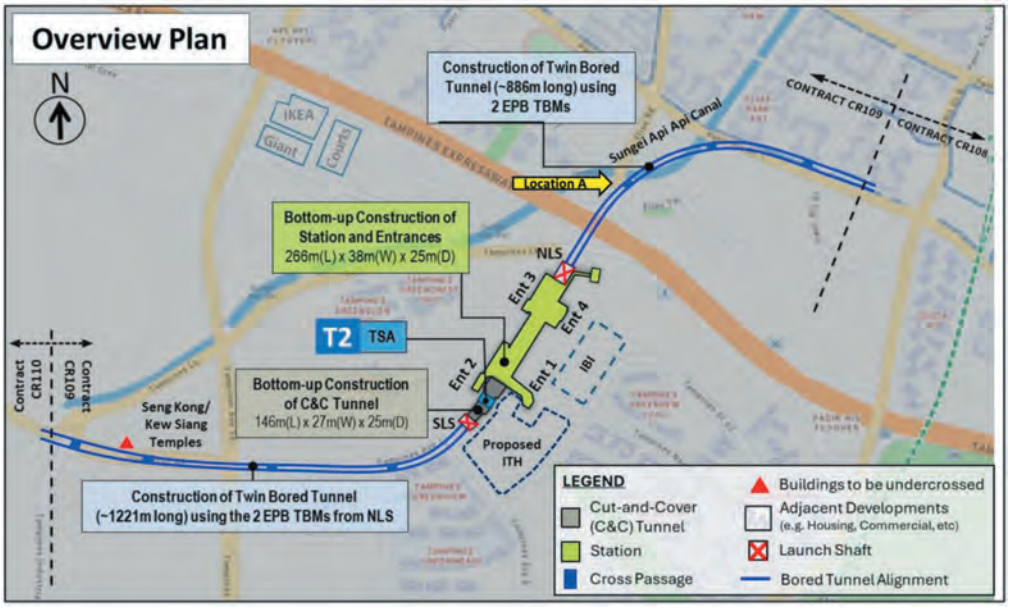


Figure 1. Project CR109 scope plan.

Two earth pressure balance (EPB) type tunnel boring machines (TBM) were deployed for the project. Both TBMs launched from the north launch shaft at Tampines North Station and constructed approximately 886m of twin bored tunnels eastward before docking at the end wall of Pasir Ris Station, denoted in Figure 1 as CR108. The TBMs were then dismantled and retrieved through the tunnels back to Tampines North Station. Each TBM was subsequently fitted with new shield skins and relaunched from the south launch shaft, undertaking the westward drives of approximately 1221m towards the end wall of the adjacent cut and cover tunnels, denoted in Figure 1 as CR110.

Along the alignment eastwards from Tampines North Station to CR108, the tunnels pass beneath a layered corridor of urban infrastructure, indicated in Figure 1 as Location A. This location presents one of the major engineering challenges on the project.

1.3 *Location A and construction worksite*

Location A comprises Elias Green road, an earth retaining wall, a pedestrian footpath approximately 2m below the road level, and the Sungei Api Api stormwater canal. The earth retaining wall which supports the level difference between the road and the footpath is founded on contiguous bored piles (CBP), and the canal is founded on bakau timber piles. The site schematic and cross section is shown in Figure 2.

During the design development phase, it was identified that the CBP piles of 1m diameter were likely to encroach into the tunnel alignment, necessitating pile extraction works. With the combination of the many closely spaced structures at Location A, this created a narrow construction worksite of approximately 4.1m width at its narrowest point. It is therefore envisaged that space-optimised engineering solutions, meticulous sequencing of construction activities and coordinated machinery movements were required to carry out the pile extraction works at Location A.

2 GEOLOGICAL CONDITIONS

The bored tunnel alignment was located in geological conditions of Kallang Formation and Old Alluvium. Old Alluvium is composed of alluvial sediments that accumulated within deep basins

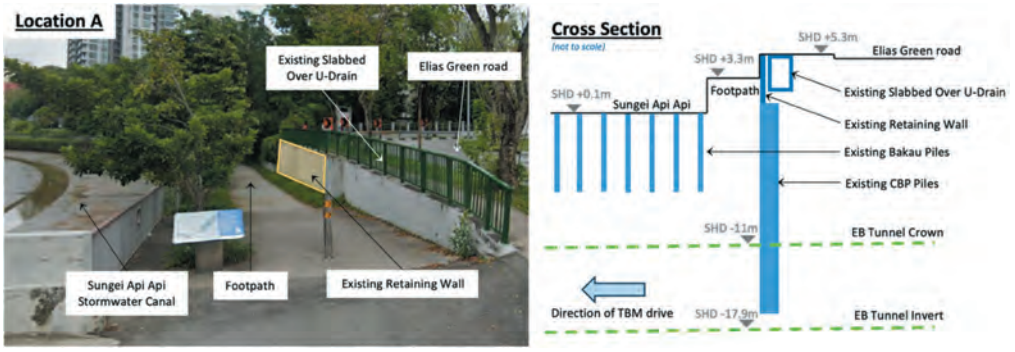


Figure 2. Site schematic and cross section at Location A.

during the late Tertiary to middle Pleistocene period, roughly 5 to 0.5 million years ago. With thicknesses than can exceed 200 metres, these deposits are typically dense and lateritic, offering uniform and stable ground conditions that are highly compatible with EPB tunnelling methods.

Old Alluvium is overlaid by Kallang Formation, a much younger layer that developed from the late Pleistocene period (0.14 million years ago) to present day. It occurs primarily along river valleys and comprises alternating layers of marine clay, sand and organic peaty soils.

At Location A, the tunnel is constructed in Old Alluvium with a thin layer of Kallang Formation comprised of fluvial sand (F1) and estuarine (E) is encountered above the tunnel alignment. The presence of Kallang Formation at this already challenging location demanded more meticulous control of face pressure and groundwater during tunnelling to prevent excessive settlement.

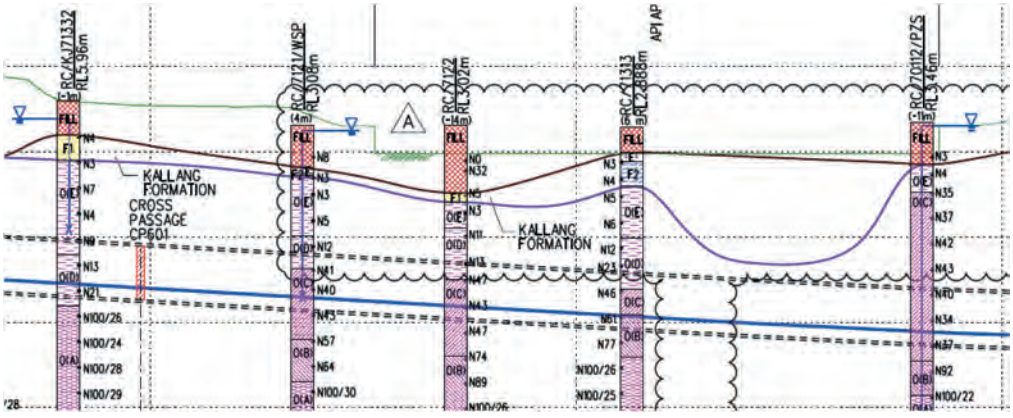


Figure 3. Geological conditions at Location A.

3 FOUNDATION INVESTIGATION WORKS

3.1 *Earth retaining wall with CBP piles*

Foundation investigation works to determine the extent of CBP pile encroachment into the tunnel alignment comprised of a combination of trenching and magnetic logging methods. Trenching provided visual confirmation of the pile positions and magnetic logging mapped the signatures of steel reinforcement within the piles.

For the magnetic logging test, four boreholes were pre-drilled within 0.5m to 1.5m from the anticipated CBP pile locations and up to a depth of 3m to 5m below the



Figure 4. Lowering of probe for magnetic logging test to determine CBP pile toe levels.

anticipated pile toe levels. After a standard penetration test (SPT) is carried out, a probe is lowered into the borehole as shown in Figure 4.

The probe is used to detect magnetic signals from any nearby magnetic materials. In the case of CBP piles, signals from any steel reinforcement will be picked up by the probe. All four points picked up magnetic signals and it was interpreted that the piles terminated at depths corresponding to below the tunnel axis, with the deepest one just 0.8m above the tunnel invert level.

3.2 *Stormwater canal with bakau timber piles*

Beyond the retaining wall, both east bound (EB) and west bound (WB) tunnels passed beneath Sungei Api Api supported on bakau timber piles. Based on design drawings, the piles are spaced 1m centre-to-centre with depths extending 9m to 15m below the cut-off level. The worst-case scenario of 15m-deep pile toe levels could potentially affect tunnelling operations.

To confirm the depth of the bakau piles, microtremor survey was conducted on the canal following the tunnel alignment. This non-intrusive foundation investigation relies on measuring the velocity and frequency of seismic vibrations to identify changes in subsoil properties. The survey points were arranged in 2m-radii pentagonal arrays comprised of 8 sensors, as shown in Figure 5. The distance between two adjacent arrays is about 1m to 10m along the centre of the EB and WB tunnel alignments to form a two-dimensional profile per tunnel. The microtremor survey interpretations showed that there is no encroachment of the bakau timber piles into both the EB and WB tunnel alignments.

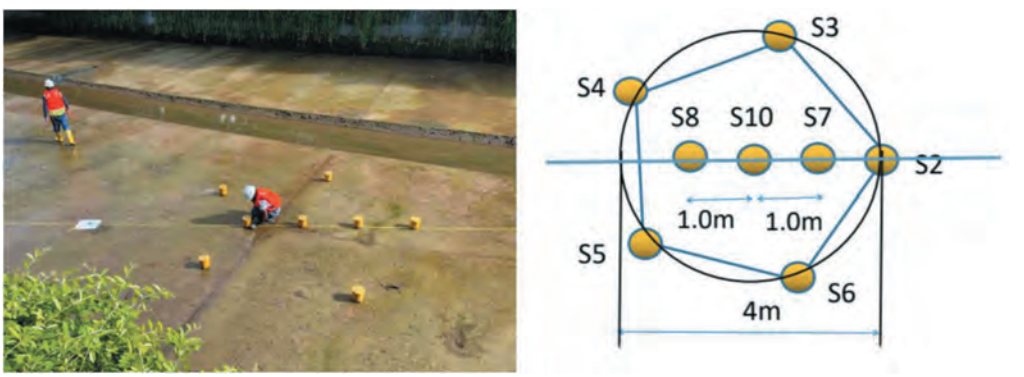


Figure 5. Configuration of one pentagonal array for microtremor survey.

4 CBP REMOVAL WORKS

Following the foundation investigation works that determined encroachment of CBP piles into the EB tunnel alignment, a series of activities were carried out to enable pile removal works.

First, an existing roadside drain running along Elias Green road was diverted and a sewer pipe within the worksite was protected. Then, a temporary earth retaining and stabilising system (ERSS) was implemented comprising of a row of soldier piles driven to an approximate 11m depth with steel lagging, with local zones strengthened with jet-grout columns to increase ground stiffness and reduce deflection during subsequent excavation. Once the ERSS was in place, the retaining wall section above the CBP piles was demolished to expose the pile heads. The layout of the CBP piles requiring removal is shown in Figure 6.

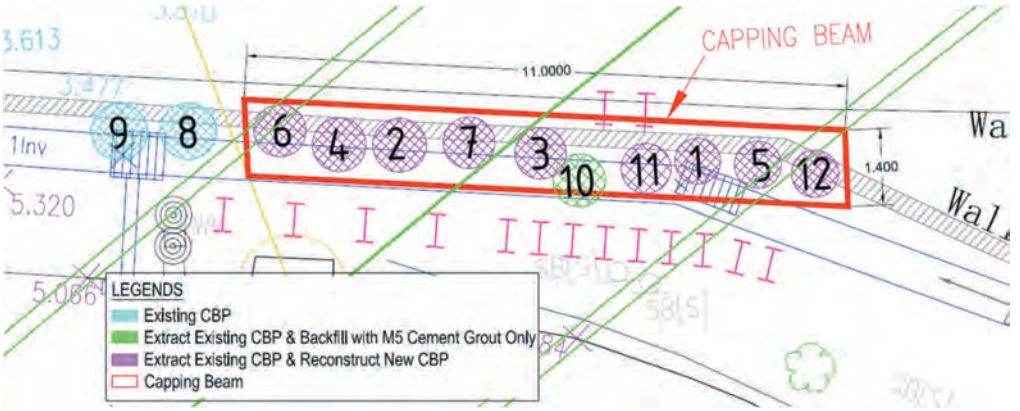


Figure 6. CBP pile layout plan.

4.1 *Pile extraction procedure*

4.1.1 *Initial method*

After exposing the pile heads, a detailed step-by-step pile extraction procedure was set up to remove the piles:

1. Drilling into the middle of the CBP pile and installation of threaded steel bar to form the connection point for the subsequent jacking frame. The threaded bar is grouted into place.
2. Drilling of a grouting hole to the pile toe level.
3. Debonding around the CBP pile perimeter by wash boring to loosen the surrounding soil and reduce shaft friction, facilitating smoother uplift during jacking operations.
4. Installation of temporary steel casing to 1m below the pile toe level to retain the surrounding ground and provide a stable working platform.
5. Hydraulic jack setup and jacking operations connected to the threaded bar. Jacking is carried out at 1m intervals with grouting carried out immediately after each interval, followed by pile hacking. The process is repeated at 1m increments until the entire pile length is removed.
6. Backfilling of the annular space and pile void with cementitious grout.

Despite the well-planned sequence, initial field implementation on the first CBP pile revealed opportunities for improvement. During the extraction works for the first pile at Pile #2, the method proved time consuming. The debonding works by wash boring took 6 days, the installation of casing by vibration method ran into significant challenges and took a total of approximately 30 days, and the jacking works took 10 days. The jacking force applied was 190T,

significantly higher than the weight of the CBP pile itself of only about 40T. The pile extraction method, specifically the debonding method, was refined with the subsequent CBP piles.

4.1.2 *Improvement to the casing installation method*

For the subsequent piles, debonding was carried out by installing a temporary rotator casing of 1.1m diameter, with a 50mm gap surrounding the 1m diameter CBP piles, to the pile toe depth. Once the temporary casing is removed, a permanent casing is installed to a depth of about 12m. The installation of the temporary casing by rotator method required a duration of 2 to 5 days, significantly faster than by vibration method which required 15 days. For Pile #7, the jacking force applied was also lower at 160T.

4.1.3 *Pressure relief methods*

This method was further refined for Pile #6, with the grouting hole wash bored after completion of debonding by rotator casing. This enabled the grouting hole to function as a pressure relief hole from the pile toe. This was successful in reducing the suction effect at the pile toe, requiring only 90T jacking force for extraction of Pile #6.

The final approach further introduced relief holes around the perimeter of the casing. This helped with the buoyancy of the pile during jacking works, further reducing the jacking force to 75T for Pile #5.

4.2 *Pile extraction performance*

The progressive refinement of the pile extraction method led to significant improvements in efficiency and control. The switch from vibratory to rotator casing installation shortened the average extraction cycle per pile, and the implementation of pressure relief holes substantially reduced the jacking forces required – from about 160T for the first pile done with rotator casing to about 75T for the final ones. Some of the equipment utilised are shown in Figure 7.



Figure 7. Drill rig for threaded bar installation, drill rig for installation of rotator casing, and jacking setup.

All affected piles were eventually extracted to 3m above the tunnel crown before the TBM advanced beneath the location. This clearance ensured that no residual obstruction remained within the tunnel face. The successful completion of these works demonstrated the effectiveness of iterative performance-based refinement adopted for the CBP pile removal process in a constrained urban environment.

5 TUNNELLING PERFORMANCE

Prior to reaching Location A, cutterhead inspections (CHI) were carried out approximately 40m before the earth retaining wall and approximately 75m before the canal for the EB and WB tunnels respectively. To further minimise impact to the surrounding structures, a volume loss limit of

0.5% was also imposed, more stringent than the authority's requirement of 1.0% in Old Alluvium geological conditions.

During construction of the EB tunnel beneath the CBP piles, the TBM maintained stable performance with minimal fluctuation in face pressures and cutterhead torque, indicating consistent TBM behaviour through the location. Metallic fragments such as two pieces of 650-660mm length rebars were discharged through the screw conveyor at separate times – one directly after undercrossing the CBP location and the second, some 20 rings later. These fragments potentially correspond to small remnants of reinforcement bars but did not have impact to TBM operational parameters and did not impede TBM advance. The settlement readings were well within allowable limits and the volume loss near the CBP location was measured at 0.25%, well within the imposed 0.5% allowance.

Subsequent construction of the EB tunnel beneath Sungei Api Api was equally uneventful. The transition from the zone beneath the earth retaining wall to the canal occurred over an approximate 5m distance, with continuous face pressure and advance rate monitoring confirming smooth excavation. Construction of the WB tunnel was also smooth with minimal deviations in TBM operational parameters. Instrumentation readings showed settlement remained well-controlled with negligible volume loss, and negligible movement to nearby structures.

Throughout the undercrossing of both structures, no anomalies were observed in the face pressure, torque or muck consistency, and instrumentations recorded minimal movements. These indicate that ground behaviour remained stable and predictable.

6 CONCLUSION

The ability to mitigate the challenges undercrossing the earth retaining wall and stormwater canal at Contract CR109 demonstrate a structured approach to tunnelling beneath existing pile-supported structures within Singapore's dense urban environment.

The pile extraction works, starting with threaded bar installation at the first CBP pile, commenced in March 2024 and the TBMs passed beneath the earth retaining wall and the canal in November 2024 and February 2025 for the WB and EB tunnels respectively. This underscores the importance of early foundation investigation. The combination of physical methods such as trial trenching and geophysical methods both intrusive and non-intrusive i.e. magnetic logging and microtremor respectively provided a clear understanding of the foundation conditions at site. This enabled identification of conflicts and the mitigation of them early in the construction stage.

The iterative refinement of pile extraction methodology from vibration-installed to rotator-installed casings and the introduction of pressure relief holes substantially improved productivity. The successful reduction of jacking forces from 190T to 75T without compromising ground stability exemplifies how data-driven refinements can yield tangible performance improvements.

During tunnelling operations, more stringent TBM operation controls were implemented. TBM performance was carefully executed based on feedback from TBM operational parameters and instrumentation readings. Successful completion of tunnelling operations validated both the design assumptions and the pile extraction approaches.

From a broader perspective, the lessons drawn – particularly the integration of early investigation, adaptive construction methodologies and stringent operational control – provide a useful reference framework for future tunnelling projects in Singapore and comparable urban settings.

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